Chapter 1 - Image Size

How much space is occupied by an image that has horizontal and vertical resolutions 1024 and 1024 and uses 3 bytes for the storage of each pixel's colour?

Answer: 1024 X 1024 X 3 bytes or 3MBytes. This is a 1 Mpixel image.

Note: Mega used here in binary (1024^2)

What if a 32bpp Z-buffer is also involved?

Answer: Add 1024 X 1024 X 4 = 4MBytes. Total 7MBytes per image.

What if a 2bpp CLUT is used?

Answer: 1024 X 1024 X .25 bytes = 256KBytes plus 4 X 3bytes = 12bytes for the CLUT.

Chapter 2 - Mathematical Curves

Give the parametric and implicit functions for a circle centered at (0,0) and with radius r=10 and hence determine the relationship of the points (9,0), (10,0) and (11,0) to the circle.

Solution:

parametric:

$x(t) = 10 \cos (2\pi t)$

$y(t) = 10 \sin (2\pi t)$

implicit:

$c(x,y) = x^2+y^2-10^2$

Hence,

$c(9,0) = -19$ inside

$c(10,0) =$ on circle

$c(11,0)=21$ outside

Chapter 2 - Finite Differences

Determine the finite differences in x and y for the line segment

$l(x,y) = 5x+6y+3$

and the circle

$c(x,y)=2x^2+3y^2-10^2$

and hence evaluate these functions on the grid

		(2,1)
(0,0)		

Solution:

$l(0,0)=3$	$c(0,0)= -100$		
$\delta_x l(x,y)=5$	$\delta_x c(x,y)=4x+2$	$\delta_x c(0,0)=2$	$\delta^2{}_x c(x,y)=4$
$\delta_y l(x,y)=6$	$\delta_y c(x,y)=6y+3$	$\delta_y c(0,0)=3$	$\delta^2{}_y c(x,y)=6$

Thus

$l(1,0)=3+5=8$ $l(2,0)=8+5=13$

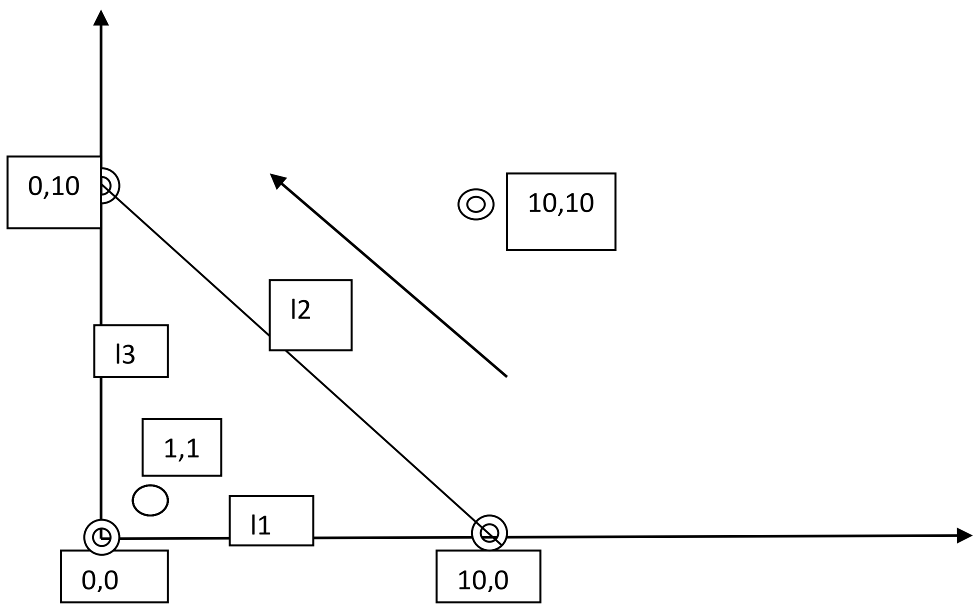
$l(0,1)=3+6=9$ $l(1,1)=9+5=14$ $l(2,1)=14+5=19$

$c(0,0)=-100$	$c(1,0)=-100+2=-98$	$\delta_x c(1,0)=2+4=6$	$c(2,0)=-98+6=-92$
$c(0,1)=-100+3=-97$	$c(1,1)=-97+2=-95$	$\delta_x c(1,1)=2+4=6$	$c(2,0)=-95+6=-89$

Chapter 2 - Sign Test for Triangle

Compute line equations making up the triangle with vertices (0,0), (10,0) and (0,10) and thus determine if the points (1,1) and (10,10) are inside the triangle.

Note: for $l(x,y)=ax+by+c$ defined between (x_1,y_1) and (x_2,y_2), $a=y_2-y_1$, $b=x_1-x_2$, $c=x_2y_1-x_1y_2$



l1: a=0, b=-10, c=0

l2: a=10, b=10, c=-100

l3: a=-10, b=0, c=0

Thus *l1(1,1)=* ***-10****, l2(1,1)=****-80****, l3(1,1)=****-10*** *Inside as all signs -ve*

l1(10,10)= ***-100****, l2(10,10)=****100****, l3(10,10)=****-100*** *Outside as signs differ*

Chapter 2 - Postfiltering Antialiasing

Assume the 3X3 subpixels values of the virtual image below and the 3x3 convolution filter shown. Compute the value of the final pixel using postfiltering. Note: you need to normalize the filter values.

Virtual Image Subpixels

8	16	8	...
16	4	8	...
2	2	4	...
...	...	...	...

Convolution Filter

1	2	1
2	4	2
1	2	1

Answer

The normalized convolution filter is:

1/16	2/16	1/16
2/16	4/16	2/16
1/16	2/16	1/16

Thus the final image pixel value corresponding to these subpixels is:

*8*1/16+16*2/16+8*1/16+16*2/16+4*4/16+8*2/16+2*1/16+2*2/16+4*1/16 = 120/16*

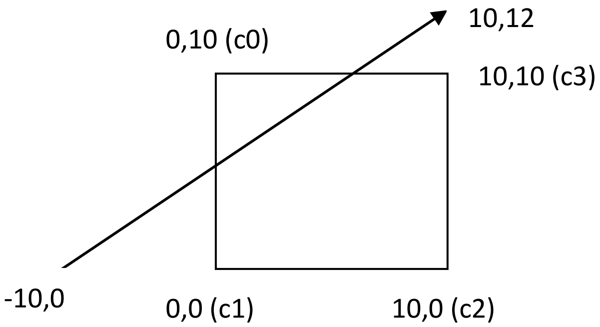
7.5	...
...	...

Chapter 2 - Scala Line Clipper

Determine the c0,c1,c2,c3 codes for the line segment and the clipping window shown below.

What do you think the code signifies?

Note: for $l(x,y)=ax+by+c$ defined between (x_1,y_1) and (x_2,y_2), $a=y_2-y_1$, $b=x_1-x_2$, $c=x_2y_1-x_1y_2$



Answer

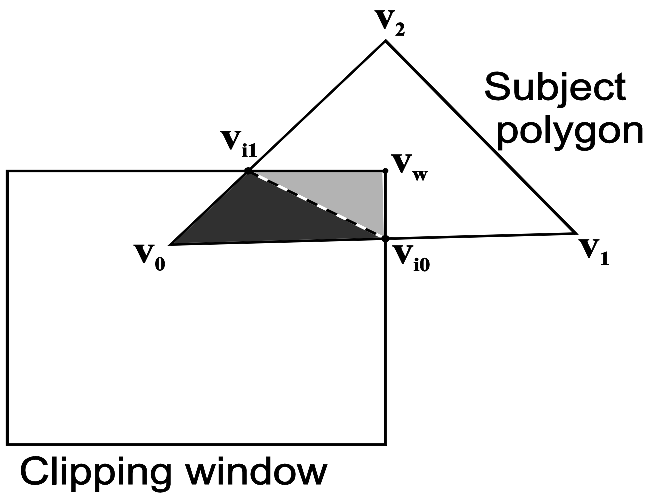
a=12 b=-20 c=120

Thus, c0=0 c1=1 c2=1 c3=1

The code 0111 therefore signifies intersection with window edges 01 and 30.

Chapter 2 - Sutherland-Hodgman Polygon Clipper

Show the successive stages of clipping of the following subject polygon against the clipping window.



Answer

Stage 0 (left): v0 v1 v2 (no clip)

Stage 1 (right): v0 vi0 vi2

Stage 2 (bottom): v0 vi0 vi2 (no clip)

Stage 3 (top): v0 vi0 vw vi1

Chapter 3 - Coordinate Systems

Use an example matrix (e.g. S(2,2)) to show the difference between applying the matrix to a column vector (as per book):

$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} m_1 & m_2 & m_3 \\ m_4 & m_5 & m_6 \\ m_7 & m_8 & m_9 \end{bmatrix} \cdot \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

or to a row vector:

$$\begin{bmatrix} p_{x'} & p_{y'} & p_{z'} \end{bmatrix} = \begin{bmatrix} p_x & p_y & p_z \end{bmatrix} \cdot \begin{bmatrix} m_1 & m_4 & m_7 \\ m_2 & m_5 & m_8 \\ m_3 & m_6 & m_9 \end{bmatrix}$$

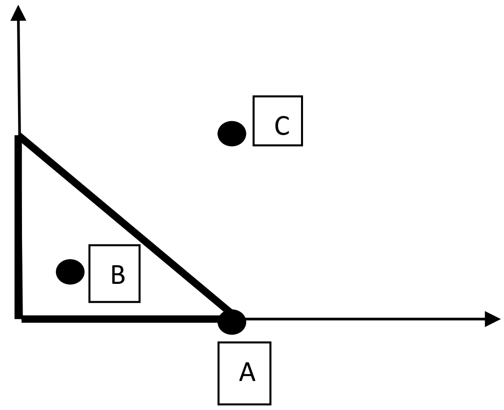
Show how the matrix would be applied to a multitude of points (making up a shape).

Chapter 3 - Affine Transformations

For the triangle with vertices (0,0) (1,0) (0,1) [affine combination], show the points that correspond to weight values A(0,1,0)   B(0.33,0.33,0.33) and C(1,1,1).

Answer

A=(1,0)   B=(0.33,0.33)   C=(1,1)



Chapter 3 - Composite Transformations

Suppose that we need to apply the sequence of transformations:

$S(3,2) \rightarrow R(90^o) \rightarrow S(2,1) \rightarrow SH_x(a) \rightarrow R(30^o)$

to a shape consisting of 10,000 points:

$[p_1 \cdots p_{10,000}]$

Estimate the number of (simple) multiplications necessary if:

(a) the matrices are directly applied $R(30^o)\ [SH_x(a)[S(2,1)\ [R(90^o)\ [S(3,2)\ \mathbf{p_i}]]]]$ i:1..10,000

(b) the matrices are first composed $M = [[[[R(30^o)\ SH_x(a)]S(2,1)]\ R(90^o)]\ S(3,2)]$ and then applied $M\ \mathbf{p_i}$ i:1..10,000

Answer

mm: 2x2matrix*matrix is 8 * mv: 2x2 and 2x1 matrix*vector is 4 *

(a) 5 mv * 10,000 = **200,000 ***

(b) 4 mm = 32 * plus 10,000 mv = 40,000 *. Total **40,032 ***

Chapter 3 - Homogeneous Coordinates

Describe the affine transformation instances $\Phi(P)=A(P)+t$ for the 2D homogeneous transformations.

Answer

Translation: $A(P)=T(d)$ & $t=0$

Scaling: $A(P)=S(s_x,s_y)$ & $t=0$

Rotation: $A(P)=R(\theta)$ & $t=0$

Shear: $A(P)=SH(\alpha)$ & $t=0$

Chapter 3 –3D Rotations

Find a memorization rule for the rotation matrices.

Answer

$$\mathbf{R}_x(\theta)=\begin{bmatrix}1 & 0 & 0 & 0\\ 0 & \cos\theta & -\sin\theta & 0\\ 0 & \sin\theta & \cos\theta & 0\\ 0 & 0 & 0 & 1\end{bmatrix} \quad \mathbf{R}_y(\theta)=\begin{bmatrix}\cos\theta & 0 & \sin\theta & 0\\ 0 & 1 & 0 & 0\\ -\sin\theta & 0 & \cos\theta & 0\\ 0 & 0 & 0 & 1\end{bmatrix} \quad \mathbf{R}_z(\theta)=\begin{bmatrix}\cos\theta & -\sin\theta & 0 & 0\\ \sin\theta & \cos\theta & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix}$$

The axis row/col has the ID data.

Ordering generally is

cosθ -sinθ

sinθcosθ

except for the Y rotation. In that case the axes ‘view’from +Y is opposite.

Chapter 4 –Properties of Transformations

Show (using a symbolic example) what generally holds for a perspective projection in relation to the ratios of distances and cross ratios.

Answer

<Draw axes, projection plane and a line segment>

Take a line segment **ab** its midpoint **m** and another point on it **c**.

Then let: [**a'b'm'c'**] = P_{PER}[**abmc**]

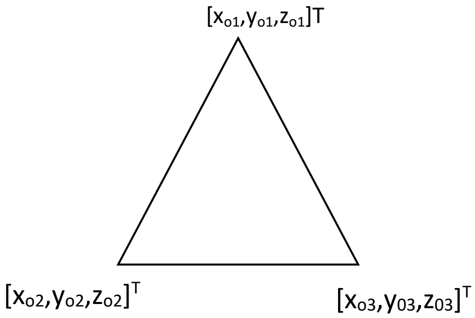
m' is not generally $\frac{a'+b'}{2}$, but $\frac{\frac{a'm'}{m'b'}}{\frac{a'c'}{c'b'}} = \frac{\frac{am}{mb}}{\frac{ac}{cb}} = \frac{1}{\frac{ac}{cb}}$

Chapter 5 –HSE: Z-buffer

Show the steps involved in converting the vertex data into depth data per pixel.

Answer

Initially, we have the data in OCS:



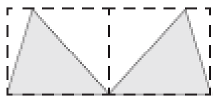
$$\begin{bmatrix} x_{o1} & x_{o2} & x_{o3} \\ y_{01} & y_{02} & y_{03} \\ z_{01} & z_{02} & z_{03} \end{bmatrix} \xrightarrow{OCS->WCS} \begin{bmatrix} x_{w1} & x_{w2} & x_{w3} \\ y_{w1} & y_{w2} & y_{w3} \\ z_{w1} & z_{w2} & z_{w3} \end{bmatrix} \xrightarrow{WCS->ECS} \begin{bmatrix} x_{e1} & x_{e2} & x_{e3} \\ y_{e1} & y_{e2} & y_{e3} \\ z_{e1} & z_{e2} & z_{e3} \end{bmatrix} \xrightarrow{ECS->NDC} \begin{bmatrix} x_{n1} & x_{n2} & x_{n3} \\ y_{n1} & y_{n2} & y_{n3} \\ z_{n1} & z_{n2} & z_{n3} \end{bmatrix}$$

$$\xrightarrow{NDC->DC} \begin{bmatrix} x_{d1} & x_{d2} & x_{d3} \\ y_{d1} & y_{d2} & y_{d3} \\ z_{d1} & z_{d2} & z_{d3} \end{bmatrix} \xrightarrow{Interpolate} z_{pixel}$$

Chapter 5 –Bounding Volumes

When considering bounding volumes (BV) for intersection tests, do we need to have a special case for neighbouring elements (e.g. adjacent triangles)?

Answer



Neighbouring elements will always have intersecting BVs. We should therefore treat them as a special case, as they will normally be considered as non-intersecting.

In the case of polygons, this can be done by disabling the BV test if 2 polygons have common vertices.

Chapter 5 –Space Subdivision

Show the quadtree that would result from the following scene.

Answer

<draw a scene in a square, and keep subdividing quad-tree style>

Chapter 9 –Scene Graphs

If

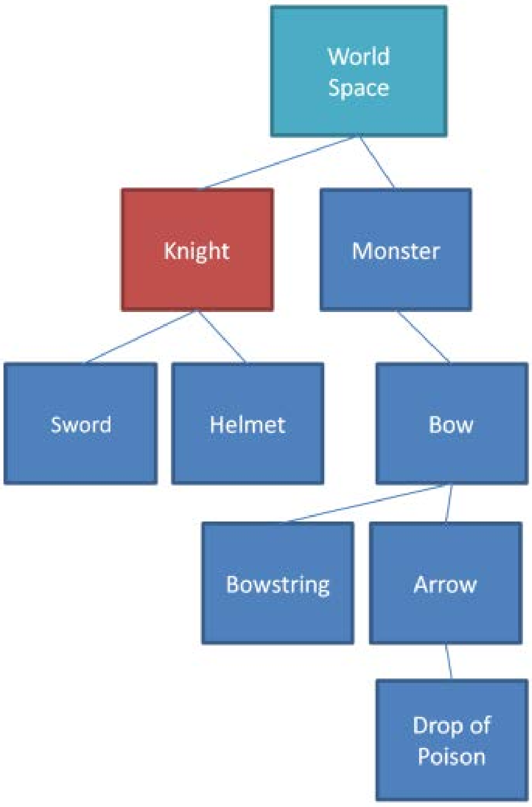
$M_{MONSTER}$ is the transformation of the Monster relative to WCS,

M_{BOW} is the transformation of the Bow relative to the Monster,

M_{KNIGHT} is the transformation of the Knight relative to WCS and

M_{SWORD} is the transformation of the Sword relative to the Knight

Then, what is the transformation of the *target* Sword relative to the *reference* Bow?



Answer

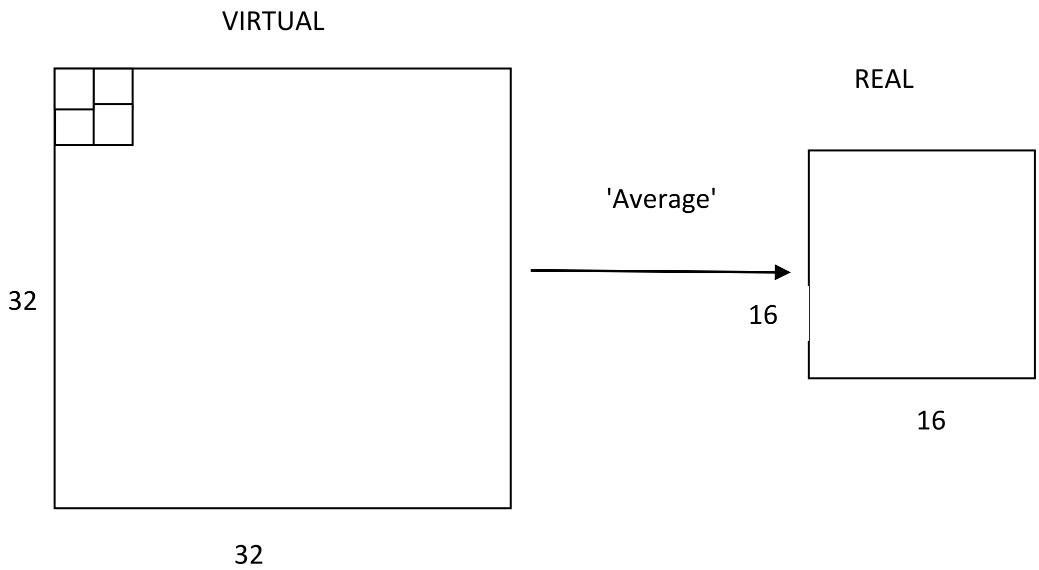
$M_{BOW}^{-1} * M_{MONSTER}^{-1} * M_{KNIGHT} * M_{SWORD}$

Chapter 11 –Anti-Aliasing vs Halftoning

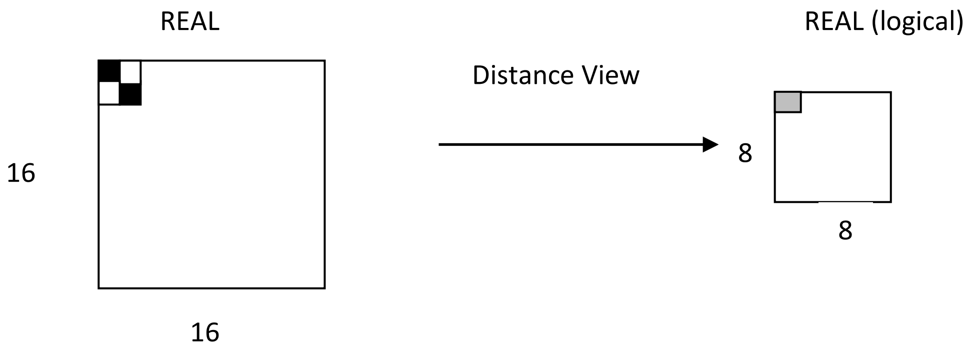
Show the difference between anti-aliasing and halftoning with a graphical example.

Answer

Anti-aliasing



Halftoning



Chapter 11 –Halftoning

Create the H_8 halftoning matrix

Answer

Use the recursive definition on the board

$$\mathbf{H}_2 = \begin{bmatrix} 0 & 2 \\ 3 & 1 \end{bmatrix}$$

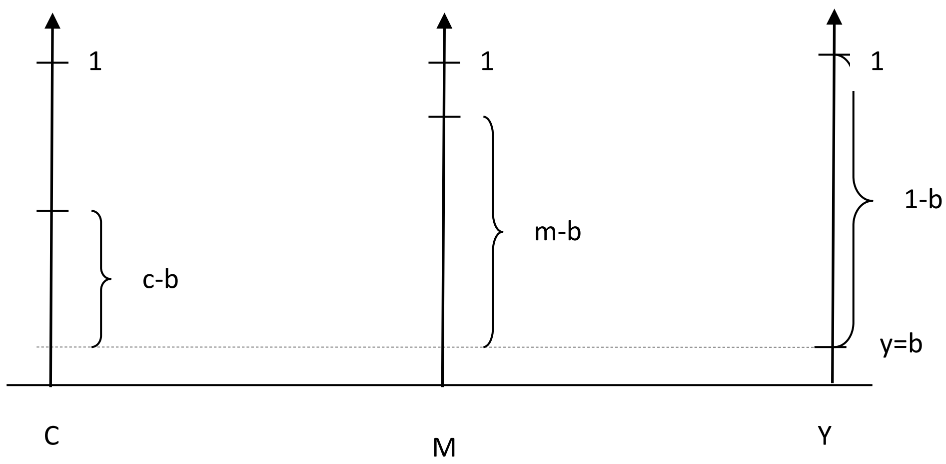
$$\mathbf{H}_\mathbf{m} = \begin{bmatrix} 4 \cdot \mathbf{H}_{\mathbf{m}/2} & 4 \cdot \mathbf{H}_{\mathbf{m}/2} + 2 \cdot \mathbf{U}_{\mathbf{m}/2} \\ 4 \cdot \mathbf{H}_{\mathbf{m}/2} + 3 \cdot \mathbf{U}_{\mathbf{m}/2} & 4 \cdot \mathbf{H}_{\mathbf{m}/2} + \mathbf{U}_{\mathbf{m}/2} \end{bmatrix}, \quad m \geq 4, \quad m = 2^k$$

Chapter 11 –CMYK

Show graphically what the CMYK colour specification represents.

Answer

K (black in Japanese, 'Kuro') represents the equal amount of CMY that is used as the common vector. One of CMY will thus be 0 and the other 2 represent the colour shift vector that must be added to reach the desired colour.



$$b = \min(c, m, y)$$

$$c' = \frac{c-b}{1-b}$$

$$m' = \frac{m-b}{1-b}$$

$$y' = \frac{y-b}{1-b}$$

Chapter 12 –Lighting

A. Name the parameters that affect lighting of a point **p** on an object, in your opinion.

B. Show which of these parameters are taken into account in the Phong model

Answer

	Phong?
Direction of light source,	Yes **l**
Distance of light source,	Yes **d**
Intensity of light source	Yes **I**
'Colour' of light source,	No
Direction of view,	Yes **v**
Normal vector at **p**,	Yes **n**
Material reflectance properties,	Yes k_a, k_d, k_s
Emission from object itself,	Yes I_e
Light interaction with other objects,	No
Medium properties	No

...

Chapter 12 –Phong model

Determine the cost of evaluating the Phong model for 1 point.

Answer

$\hat{\boldsymbol{n}}$: 3 additions (interpolation) and [3* 3/ 1√ 2+]=K (normalization)

$\hat{\boldsymbol{l}}$: 3 subtractions (l-p) and K (normalization)

$\hat{\boldsymbol{v}}$: 3 subtractions (v-p) and K (normalization)

$\hat{\boldsymbol{r}}$ / $\hat{\boldsymbol{h}}$: [6* 3+] ($\vec{\boldsymbol{r}}$) or [3+] ($\vec{\boldsymbol{h}}$) and K (normalization)

$\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{l}}$: 3* 2+

$(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{h}})^m$: (3+m)* 2+

$I = I_e + I_a k_a + I_i(k_d(\hat{\mathbf{n}} \cdot \hat{\mathbf{l}}) + k_s(\hat{\mathbf{n}} \cdot \hat{\mathbf{h}})^n)$: 4* 3+

TOTAL ~12* ~20+ and 4 [3* 3/ 1√ 2+] normalizations

Chapter 12 –Phong vs Gouraud algorithm

Compare the cost per point of the Phong and Gouraud algorithms for the Phong model.

Answer

Phong algorithm

As above ~12* ~20+ and 4 [3* 3/ 1√ 2+]

Gouraud algorithm

1) Evaluation of Phong model at 3 triangle vertices: 3*{ above less normal computation} or 3*{~12* ~17+ and 3 [3* 3/ 1√ 2+]}

2) Intensity interpolation from point to point: ~1+

Difference per point to Phong algorithm is quite large, as step 1) is shared among the many interior points of the triangle.